RESEARCH DATA MANAGEMENT · MEPI 2026-27

REDCap Part 2
Advanced features

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Today

Longitudinal projects
Arms, events, repeating
Surveys
Data Access Groups
Field embedding and piping
Action tags, smart variables
Class demo: TBCARE
Homework: TUMAINI

Slides, forms, XML and homework: **bosangir.github.io/REDCapMepi**

You are the data manager

1. **Wrote the DMP**
2. **Chose tools and methods**
3. **Built one instrument**
4. **Now: a real study**

Ask the class: what does a real study have?

Several **visits** for the same person
Several **arms** or treatment groups
Several **sites** and many staff
Different **forms** at different visits
Visits or forms that **repeat** (unscheduled visits, adverse events)
Forms filled in by the **participant** (surveys)

One form per study is not enough: you need a **longitudinal** project.

DMP: Data Management Plan

TOPIC 1 OF 6

Longitudinal projects

1. **Longitudinal projects**
2. Arms, events and repeating
3. Surveys
4. Multi-site: Data Access Groups
5. Smarter forms: embedding, piping, action tags
6. Class demo and homework

Longitudinal projects: when and why

Use it for

Cohorts and clinical trials

Multi-arm studies

Repeated questionnaires (every 3 months)

Long follow-up with fixed visits

Advantages

Build a form once, reuse it at every visit

Visit schedule and calendar

Easy to add events later

One record per participant

Costs

More complex to set up

Large record status dashboard

Logic must name the event

Mistakes in events are hard to fix after Production

Draw the **schedule of assessments** on paper before you open REDCap.

How REDCap organises a longitudinal project

PROJECT · TBCARE

ARM 1 · drug-susceptible TB

Baseline	Month 2	Month 6	Month 12
Screening	Clinical	Clinical	Clinical
Enrolment	Laboratory	Laboratory	Laboratory
Clinical			
Laboratory			

ARM 2 · drug-resistant TB

Baseline	Month 1	Month 2	… Month 12
Screening	Clinical	Clinical	Clinical
Enrolment	Laboratory	Laboratory	Laboratory
Clinical			
Laboratory			

Arm: a group with its own schedule · **Event**: a visit or time point · **Instrument**: a form, reused across events

> A record lives in **one arm**. The same instrument can appear in many events.

REDCap: Research Electronic Data Capture · TB: tuberculosis

Setting up a longitudinal project, step by step

1. **Design all instruments**
2. **Project Setup: enable longitudinal**
3. **Define arms and events**
4. **Designate instruments to events**
5. **Set repeating events and instruments**
6. **Test with fake records**
7. **Move to Production**
8. **Collect data; change via Draft Mode**

Shortcut for today: one **project XML** file holds all of steps 1 to 5.

XML: Extensible Markup Language

TOPIC 2 OF 6

Arms, events and repeating

1. Longitudinal projects
2. **Arms, events and repeating**
3. Surveys
4. Multi-site: Data Access Groups
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Arms and events

Event name	Arm	Unique event name	Day offset	Window
Baseline	1	`baseline_arm_1`	0	0
Month 2	1	`month_2_arm_1`	60	±7
Month 6	1	`month_6_arm_1`	180	±14
Baseline	2	`baseline_arm_2`	0	0
Month 1	2	`month_1_arm_2`	30	±7

Remember

Unique event names are made by REDCap: you cannot type them

Rename events **before** data entry

Arm number is the identifier; the arm name can change

Day offset and window drive the scheduler

Unique event names (**baseline_arm_1**) are what logic, exports and R see.

Designate instruments for my events

Instrument	Day 0	M3	M6	M9	M12	Unsched.	Missed
SELIG	✓						
SEGE	✓						
FSX (survey)	✓	✓	✓	✓	✓	✓	
FGE		✓	✓	✓	✓	✓	
MV							✓
LOC	✓						

Tip

Download the grid as CSV

Upload it in another project

Untick only if no data

⚠ Unticking a form that already has data **hides** that data. Plan first.

CSV: comma-separated values · SELIG: screening and eligibility form · FSX: sexual history form · FGE: follow-up genital examination form · MV: missed visit form

Repeating events and repeating instruments

Repeating event

The **whole visit** repeats, with all its forms

Example: **Unscheduled** visit (TBCARE, TUMAINI)

Instance 1, 2, 3 … of the event

Export: redcap_repeat_instance filled, redcap_repeat_instrument empty

Repeating instrument

One form repeats inside one event

Example: **adverse events**, medicines, missed visits

Custom label pipes a value: [ae_term]

Export: both redcap_repeat_instrument and redcap_repeat_instance

Project Setup > Enable optional modules > **Repeating instruments and events**.

Data entry, scheduling and the calendar

Record status dashboard

Choose the arm when you add a record

Rows: records; columns: events

Colours: incomplete, unverified, complete

Filter by arm or by DAG

Scheduling

Enable in Project Setup

Generates visit dates from Day 0

Uses day offset and window

Shows when a visit is out of window

Calendar

Per participant or for the whole project

Day, week, month views

Status: scheduled, completed, cancelled, no show

Export to Outlook or Google Calendar

Calendar rights are a separate **user right**. Give them to site staff who plan visits.

DAG: Data Access Group

Branching logic in a longitudinal project

CLASSIC: one form, one value

```
[age] >= 18

[hiv_status] = '1'
```

LONGITUDINAL: name the event

```
[baseline_arm_1][age] >= 18

[first-event-name][cons_seq] = '1'

[event-name] = 'unscheduled_arm_1'
```

Read it as

[baseline_arm_1][age]: age at baseline of arm 1
[first-event-name][x]: x at the first event of **this record's** arm
[event-name]: the event you are in now
Without an event, REDCap uses the current event

Prefer **smart variables**: one rule then works in every arm.

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Surveys

1. Longitudinal projects
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Surveys: the participant fills in the form

	Public survey link	Participant list (personal invitation)
Who	Anyone with the link	A named participant with an e-mail or phone
Identifiable?	Anonymous unless you ask	Yes: linked to the record
Times filled in	Many times	Once per invitation
Where	First instrument of the first event only	Any instrument, any event
Tracking	No	Who responded, reminders
Use for	Recruitment, screening	Follow-up questionnaires



Or open it at the visit: **Open survey** on a tablet, as in TBCARE and TUMAINI.

Setting up a survey

1. **Project Setup: enable surveys**
2. **Build the instrument**
3. **Online Designer: Enable as survey**
4. **Survey settings: texts, save and return**
5. **Distribute: link, participant list or Open survey**

✓ Good survey settings

Instructions that pipe: [event-label]
Save and return later for long forms
Hide fields for staff with @HIDDEN-SURVEY

⚠ Switching a survey off

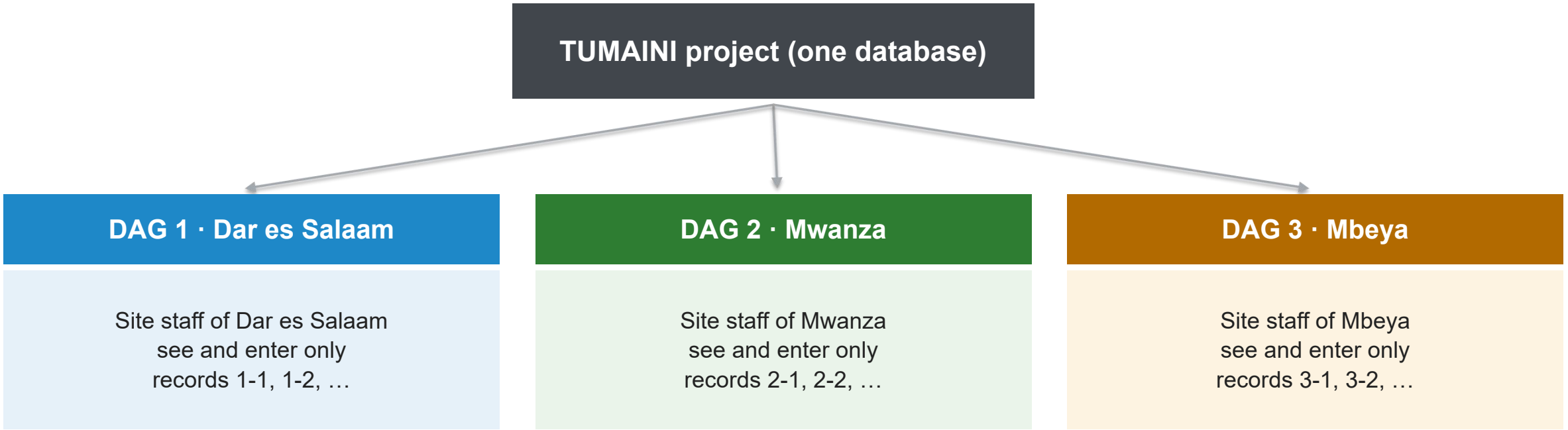
Survey settings > **Survey status: offline**
Do **not** use Delete survey settings

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Multi-site: Data Access Groups

1. Longitudinal projects
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3. Surveys
4. **Multi-site: Data Access Groups**
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Data Access Groups: one project, many sites



Data managers and monitors are in **no DAG**: they see all three sites.

> Same forms, same codes, one export: the **pooled analysis** is easy.

DAG: Data Access Group

Setting up Data Access Groups

1. Applications > Data Access Groups: create one per site
2. User Rights: assign each site user to their DAG
3. Give the role **no** Data Access Groups right
4. Turn on record auto-numbering: IDs get the DAG prefix (2-14)
5. Import: column redcap_data_access_group = unique name
6. Reports and exports: filter or add the DAG column

IMPORT FILE (CSV)

```
ptid,redcap_event_name,
redcap_data_access_group,scr_date
1-1,day_0_arm_1,dar_es_salaam,…
2-1,day_0_arm_2,mwanza,…
3-1,day_0_arm_3,mbeya,…
```

DAG Switcher

A monitor in several sites switches between them without a new account

⚠ Test it: log in as a site user and check that the other sites **are invisible**.

DAG: Data Access Group · CSV: comma-separated values

Data Access Groups or separate projects?

	Data Access Groups	One project per site
Forms and codes	**Identical by design**	Can drift apart
Changes	Made once	Repeated in every project
Pooled analysis	**One export**	Merge files, check codes
Site sees others?	No	No
Choose it when	Same protocol at all sites	Different forms, approvals or servers



Homework: TUMAINI uses **three DAGs**, one per site, one per group member.

DAGs: Data Access Groups

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Smarter forms: embedding, piping, action tags

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Field embedding: put fields where you want them

LABEL OF A DESCRIPTIVE FIELD

```
<table>
 <tr><th>Weight (kg)</th><th>Height (cm)</th></tr>
 <tr><td>{weight_kg}</td><td>{height_cm}</td></tr>
</table>
```

REDCap Clinical assessment

Weight (kg)	Height (cm)
[64.5]	[171]

- Same instrument only (same survey page)
- Works in labels, field notes, section headers and choice labels
- Use it for grids: vital signs, consent statements, test and result

> Write **{variable}** in curly brackets. The field moves; its rules stay.

Piping: reuse an answer in the text

You write	REDCap shows
`Of these [sex_n] times, how many with a condom?`	Of these 6 times, how many with a condom?
`Is [ae_term] related to treatment?`	Is Skin rash related to treatment?
`[sex]  /  [sex:value]`	Female / 2
`Dear [first-event-name][first_name],`	Dear Grace,
`This is your [event-label] questionnaire.`	This is your Month 6 questionnaire.

> Piping uses **square brackets**; embedding uses **curly brackets**.

Action tags: behaviour in one word

Action tag	What it does	Example in our studies
`@TODAY`	Fills in today's date	Visit date
`@HIDDEN-SURVEY`	Hidden on the survey, visible to staff	FSX date
`@READONLY`	Shown, cannot be changed	Eligibility text
`@CHARLIMIT=n`	At most n characters	Initials, phone
`@CALCTEXT(…)`	A calculation that returns text	ELIGIBLE / NOT ELIGIBLE
`@NONEOFTHEABOVE`	Unticks the others when chosen	Symptoms: None
`@HIDECHOICE='3'`	Hides a choice, keeps old data	Retired answer option
`@LATITUDE` `@LONGITUDE`	Captures the GPS position	Home position (LOC)
`@DEFAULT='…'`	A starting value, can pipe	Site from the DAG

> Write action tags in the **Field Annotation** box. Several can be combined.

FSX: sexual history form · DAG: Data Access Group

Smart variables and special functions

Smart variable	Gives
`[record-name]`	The record ID
`[event-name] / [event-label]`	Current event (unique / label)
`[first-event-name]`	First event of the record's arm
`[arm-number]`	Arm of the record
`[record-dag-name]`	The record's DAG
`[user-name]`	Who is logged in

Function	Example
`datediff()`	`rounddown(datediff([dob], [scr_date], 'y'), 0)`
`if()`	`if([qtcf] > 500, 1, 0)`
`round()`	`round([weight]/([height]/100)^2, 1)`
`sum(), mean()`	`sum([q1], [q2], [q3])`
`isblankormissingcode()`	`isblankormissingcode([dob])`
`concat()`	`concat([first_name], ' ', [last_name])`

✓ Test every calculation with a record where you **know the answer**.

ID: identifier · DAG: Data Access Group

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Class demo and homework

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6. **Class demo and homework**

Before the class: create the TBCARE demo project

1. Log in at redcap.uantwerpen.be
2. New Project. Title: **TBCARE demo - your student number**
3. Purpose: Practice / Just for fun
4. Option: **Upload a REDCap project XML file**
5. Choose TBCARE_2026_REDCap_project.xml
6. Create Project

What you get

11 instruments, 77 fields
2 arms, 15 events
Repeating event and instruments
2 DAGs: Antwerp, Mbarara
2 surveys
Embedding, piping, action tags

bosangir.github.io/REDCapMepi/coursework

⚠ Do it **before the class**: we will not wait for uploads.

REDCap: Research Electronic Data Capture · XML: Extensible Markup Language · DAGs: Data Access Groups

Homework: build the TUMAINI multi-centre study

Part	Task	Points
A	Project, users and roles	5
B	Six instruments (five CRFs, one of your own)	25
C	Three arms, events, form mapping	15
D	Repeating event and instrument	10
E	Three sites: Data Access Groups	15
F	Embedding, piping, action tags, survey	20
G	Test, Production, mobile app	10

Deadline

Sunday 13 December 2026
11:00 Brussels time

Hand in on Blackboard:
project XML, data dictionary,
codebook, report (max. 6 pages)

Everything you need: **bosangir.github.io/REDCapMepi/homework**

XML: Extensible Markup Language · CRFs: case report forms

Resources

Course website

Slides and course work
TBCARE XML and CRFs
Homework and TUMAINI CRFs
Quick reference (PDF)

bosangir.github.io/REDCapMepi

Inside REDCap

Online Designer: Action Tags, Piping, Smart Variables, Field Embedding, Special Functions
Help & FAQ
Training videos

Videos

projectredcap.org/resources/videos
Longitudinal projects
Surveys
Data Access Groups

Questions about the homework: in class or on the **Blackboard discussion board**.

XML: Extensible Markup Language · CRFs: case report forms · PDF: Portable Document Format · REDCap: Research Electronic Data Capture